NEBIUS × NVIDIA GLOBAL AI HACKATHON · BEST APPS AND AGENTS

Northbridge Brief

A quiet household agent. Routine bills are handled silently. Real decisions get one short brief with cited web sources.

Nebius Token Factory | Nemotron 3 Nano · Super · Ultra | Tavily search + extract | Next.js / TypeScript | Vercel

Demo household: Avery & Morgan (pseudonyms) · github.com/dl88jing/northbridge-nebius-brief

THE PROBLEM

Bills hide duplicates, price bumps and surprises

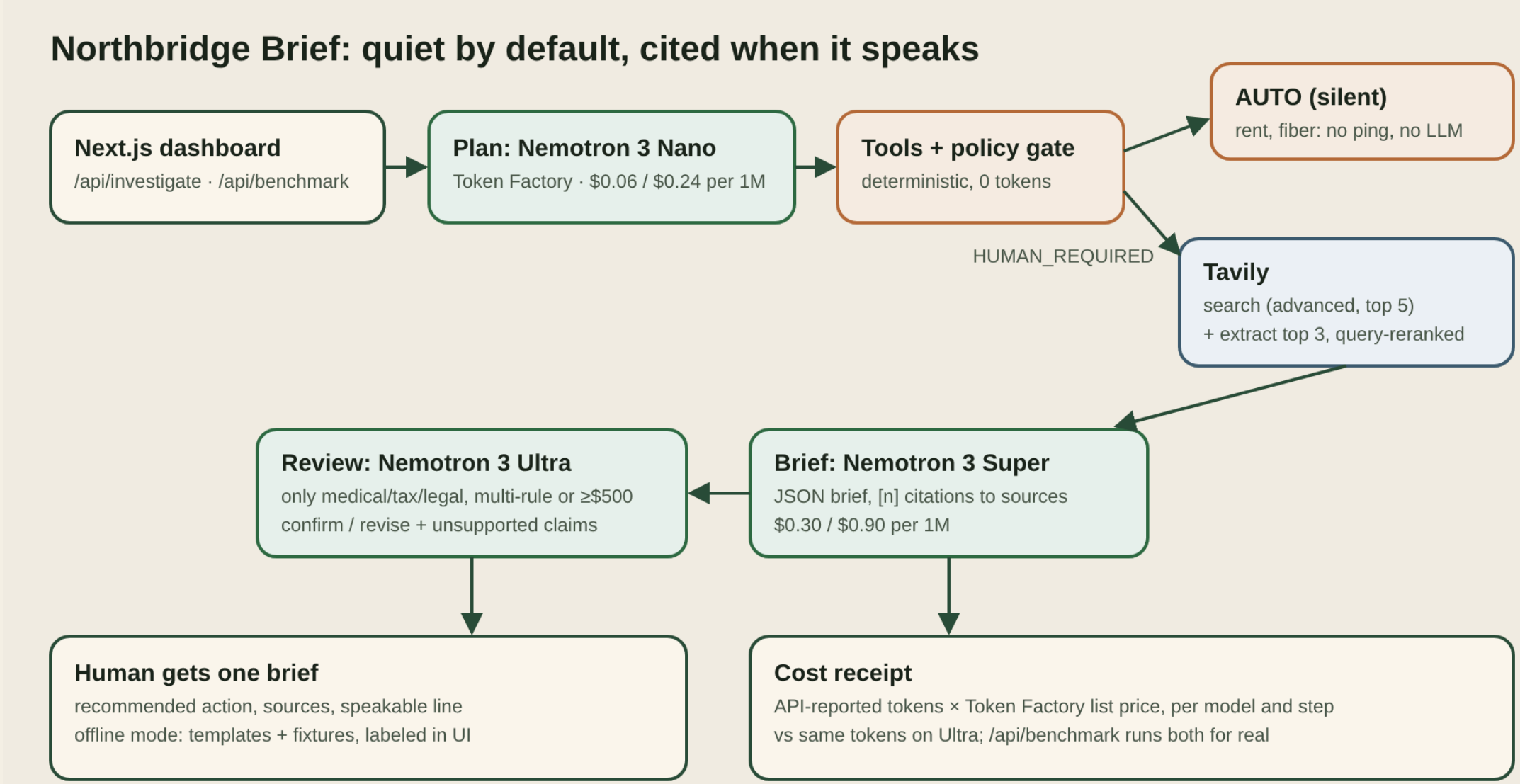
- Checking every bill is a chore, so people stop checking.
- Finance chatbots are noisy: they ping about everything.
- When something is wrong you still have to research what changed.

THE IDEA

Quiet by default, cited when it speaks

- Deterministic policy gate decides AUTO vs HUMAN_REQUIRED. No LLM, no tokens.
- Escalations get Tavily research: search, then extract the top pages.
- Nemotron Super writes the brief with [n] citations.
- Nemotron Ultra reviews only high-stakes briefs (medical, multi-rule, ≥$500).

ARCHITECTURE



MODEL ROUTING

Right-size every step

Step	Runs on	List price per 1M (in / out)
Policy gate	Rules	$0
Plan	Nemotron 3 Nano	$0.06 / $0.24
Brief	Nemotron 3 Super	$0.30 / $0.90
Escalation review	Nemotron 3 Ultra	$1.00 / $3.00

Each step in the UI shows its model, API-reported tokens, latency and cost. The app also compares against the same run on Ultra only. Prices: Token Factory public catalog, 2026-10-25.

HONEST STATUS

What is real today

- Agent loop, policy gate, UI, trace, and tests: real and running.
- Live Token Factory + Tavily path is implemented and covered by tests with mocked APIs.
- Without keys, the deployed demo runs offline. A banner says so, and no costs are shown.
- Limits: fixture household data, no bank connection, no payments made.

TRY IT

Run it in 30 seconds

Live: northbridge-nebius-brief.vercel.app

Code: github.com/dl88jing/northbridge-nebius-brief

```
git clone … && ./demo.sh · npm test
```